LOGARITHMIC EQUATIONS

Domain restrictions

Every argument of every logarithm in the original equation must be positive.

DEFINED WHEN

$\log_a(f(x))$ is defined

HOLDS WHEN $f(x) > 0$, for every logarithm in the equation

✓ Admissible

$x > 5$ for $\log_3(x-5) + \log_3(x+1) = 2$

$x - 5 > 0$ gives $x > 5$ and $x + 1 > 0$ gives $x > -1$; the intersection is $x > 5$.

$x = 2 + 3\sqrt{2} \approx 6.24$

Satisfies $x > 5$: the solution.

BOUNDARY

✗ Rejected

$x = 2 - 3\sqrt{2} \approx -2.24$

A root of $x^2 - 4x - 14 = 0$, but it fails $x > 5$: an argument would be negative.

ORDER OF WORK

before Identify the restrictions before solving: condense to $\log_3((x-5)(x+1)) = 2$, solve $(x-5)(x+1) = 9$, then test each root against $x > 5$.